

Orchestrator:

- Nasreddine Ouchene

Presentatie:

- Rosalie Post
- Mark Bongers
- (+ everyone contributes parts)

report:

- Rosalie Post
- Mark Bongers
- (+ everyone contributes parts)

Deadlines:

- Donderdag 8 oktober: Code af + bespreken in meeting
- Woensdag 14 oktober: report + presentatie af
- Oefenpresentatie: Thursday October 15
- Presentatie + report: **October 16, (ochtend presenteren en 15h bestanden inleveren)**

Vascular model: 0012_H_AO_H (als t kan)



CORE WORK

-

Fit local closures to reduce the residual.
Compare a 0D/1D model with the 3D solution.

Aan het spreken: Ibtissame E

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[illegible]

Main structure

1. Research question and hypotheses

Main question: Along a realistic 3D vessel (with bends and stenoses), how well do the assumptions behind the 1D blood-flow equations hold, and where do they break down?

Hypotheses per assumption

#	Assumption	Test it with
A1	Flow is mainly axial (small radial and secondary velocity)	Ratio of in-plane to axial velocity per slice
A2	Pressure is uniform over a cross-section	Map of cross-sectional pressure non-uniformity
A3	Velocity profile has a fixed shape, so α is constant	Measured $\alpha(x,t)$ against reference profiles
A4	Wall friction follows the assumed profile, $K_R = 2(\zeta+2)\pi v$	Measured wall shear against model friction
A5	Mass is conserved along the axis	$Q(x,t)$ consistency, and $\partial A/\partial t$ if walls are compliant
A6	The whole balance holds	Momentum-equation residual along x

Pipeline

1. Load the 3D data, extract and resample the centerline, use the last cardiac cycle
2. Slice perpendicular to the centerline. Per slice, compute A , Q , mean P , α , wall shear and in-plane velocity
3. Mass balance: check $Q(x,t)$ is consistent along the vessel. This also validates the slicing
4. Compute Womersley, Reynolds and Dean numbers along the vessel
5. Pressure non-uniformity: $(P_{\max} - P_{\min})$, normalized by $\frac{1}{2}\rho U^2$ and by the axial pressure drop
6. Compare measured α and friction with the Poiseuille, $\zeta = 9$ and Womersley reference profiles
7. Momentum residual: evaluate each 1D term along x , once with the model assumptions and once with the measured α and friction. The difference shows which assumption causes the error
8. Validity map: mark where the residual or deviations exceed a threshold (for example 10%), and link those regions to curvature, stenoses and $Wo/Re/De$

Report and presentation outline

- Abstract

- **Introduction:** why 1D models are used; the question
- **Methods:** 1D equations and their assumptions, reference profiles, slicing and integration, residual definition, verification case
- **Results:** mass balance $\rightarrow$ $W_o/Re/De$ along the vessel $\rightarrow$ pressure non-uniformity $\rightarrow$ α and friction $\rightarrow$ residual $\rightarrow$ validity map
- **Discussion:** which assumption fails where and why, sensitivity to slice spacing and thresholds, limitations (rigid walls, a single geometry), implications for 1D modelling (for example, local loss models at stenoses)
- **Appendices:** code, "who did what" table, authorship statement